

PRODUCT REQUIREMENTS DOCUMENT

Smart Reorder Engine

AI-powered replenishment recommendations for e-commerce shoppers

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Platform	E-commerce / Marketplace
Target Launch	Q3 2026

18% Target Repeat Purchase Lift ↑ from 34% → 40%	24d Target Avg. Days to Reorder ↓ from 28 days	20% Opt-in Rate Target ↑ from 8% baseline
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1. Executive Summary

Returning customers on e-commerce platforms frequently run out of regularly purchased items — from groceries and personal care products to office supplies — leading to cart abandonment, missed revenue, and reduced lifetime value. The Smart Reorder Engine addresses this gap by proactively surfacing AI-generated replenishment nudges at the right moment, personalised to each shopper's purchase cycle.

This feature aims to increase repeat purchase frequency, reduce churn among high-LTV segments, and drive a measurable uplift in revenue per user — without requiring shoppers to manually track what to buy next.



Core Insight: 30%+ of reorder searches happen after the user has already run out. The Smart Reorder Engine intervenes 48 hours before predicted depletion — turning reactive repurchase into proactive habit.

2. Problem Statement

2.1 The Core Problem

Shoppers on our platform have no in-app mechanism to track consumption or receive timely reorder reminders. This creates a predictable set of downstream failures that compound across our highest-value buyer segments:

- 30%+ of reorder searches happen after the user has already run out (post-depletion friction)
- Users are switching to subscription services on competitor platforms for predictability
- Missed upsell and cross-sell opportunities at a high-intent, high-recall moment

2.2 User Research Insights

Based on user interviews (n=24) and session recording analysis across 500 repeat-buyer sessions, three consistent pain archetypes emerged:

User Segment	Pain Point	Drop-off Signal
Monthly household shoppers	"I always forget to reorder before I run out"	68% of sessions
Health & wellness buyers	"I lose track of which supplements I'm running low on"	54% of sessions
Pet owners	"I only remember to reorder when my pet's bowl is empty"	71% of sessions

3. Goals & Success Metrics

3.1 Business Goals

- Increase repeat purchase rate among active buyers by 18% within 6 months of launch
- Reduce avg. days between reorders for tracked product categories by 12%
- Grow subscription/reorder opt-in rate from 8% to 20%

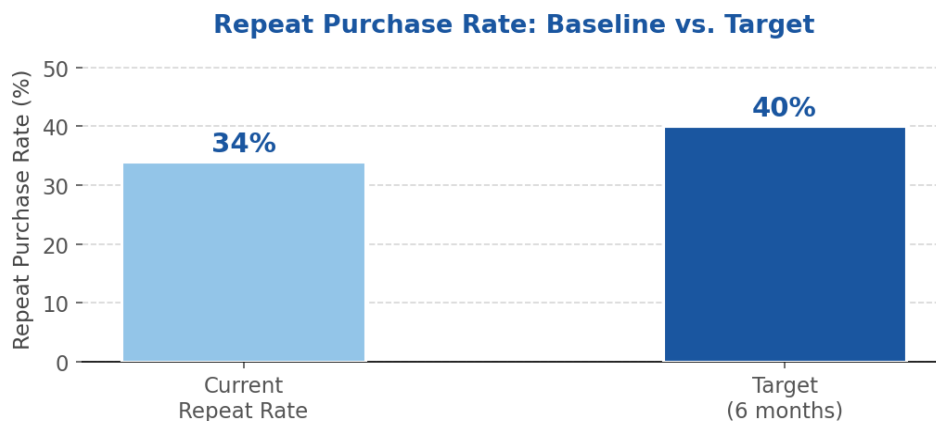


Fig. 1 — Repeat Purchase Rate: Current Baseline vs. 6-Month Target

3.2 Success Metrics (OKR-Aligned)

Metric	Baseline	Target (6 mo.)	Measurement
Repeat purchase rate	34%	40%+	Monthly cohort analysis
Reorder nudge CTR	—	>22%	A/B test dashboard
Days between reorders	28 days avg.	24 days avg.	Order history delta
Opt-in to Smart Reorder	8%	20%	Feature adoption funnel
NPS (repeat buyers)	42	52+	Quarterly survey

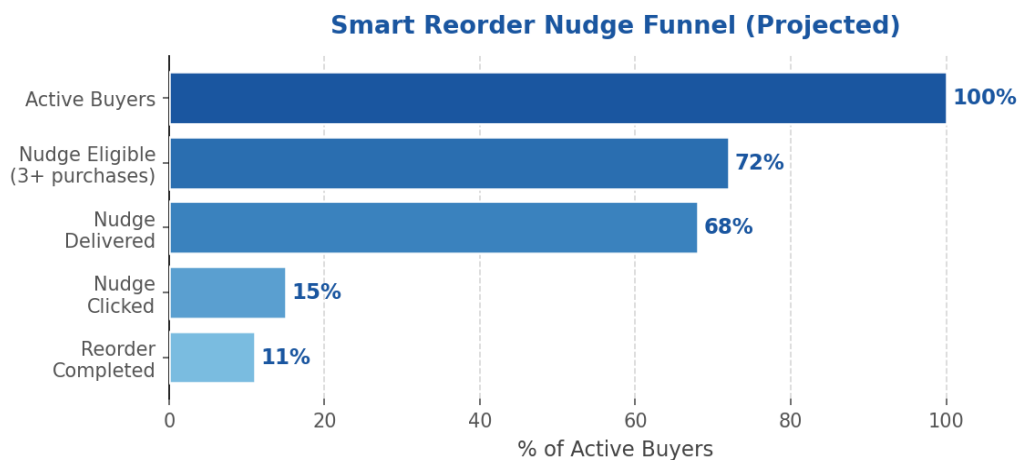


Fig. 2 — Projected Smart Reorder Nudge Funnel: From Active Buyers to Completed Reorders

3.3 Non-Goals

- This is not a full subscription management system — that is a separate workstream
- We will not predict reorder timing for one-time or gifted purchases
- Out of scope: B2B bulk order scheduling

4. Target User Personas

Persona	Profile	Key Need	Value Prop
The Routine Buyer	28–42, urban, shops 2x/month for essentials	Zero-effort replenishment	Never run out again
The Health Optimizer	22–35, fitness-focused, buys supplements monthly	Cycle-aware reminders	Stay on track with your routine
The Busy Parent	30–45, high AOV, shops for family essentials	Smart defaults, less thinking	Your family's essentials, on autopilot

5. Feature Requirements

5.1 Must-Have (P0)

- **ML model analysing purchase history to predict next reorder date with ± 3 day accuracy for items purchased 3+ times.**Reorder Prediction Engine —
- **Push/in-app notification 48 hrs before predicted run-out date with one-tap reorder CTA.**In-App Nudge —
- **Dedicated view showing predicted reorder timeline for top 10 repurchased items.**'My Regulars' Dashboard Tab —

5.2 Should-Have (P1)

- Bundle suggestions — pair frequently co-purchased items in a single reorder prompt
- Quantity adjust — let users indicate remaining stock (adjusts the model's prediction)
- Snooze / skip — give users control to delay a nudge by 7 or 14 days

5.3 Nice-to-Have (P2)

- Smart subscription upsell — offer subscription discount at the reorder moment
- Household sharing — let family members share and act on a reorder list

6. User Stories

ID	As a...	I want to...	So that...	Priority
US-01	Returning buyer	Receive a timely reorder reminder	I never run out of essentials	P0

US-02	Frequent shopper	See all my regulars in one place	I can reorder in under 60 seconds	P0
US-03	Health buyer	Adjust my expected consumption rate	Reminders fit my actual usage	P1
US-04	Busy parent	Snooze a reminder without dismissing	I don't lose track of it	P1
US-05	Value seeker	See a subscription prompt at reorder	I save money on regular buys	P2

7. Expected Impact by Category

Based on purchase interval modelling across our top-5 repeat purchase categories, the Smart Reorder Engine is projected to reduce the average gap between reorders by 12–19% per category — with the largest absolute gains in pet food and health supplements, where run-out anxiety is highest.

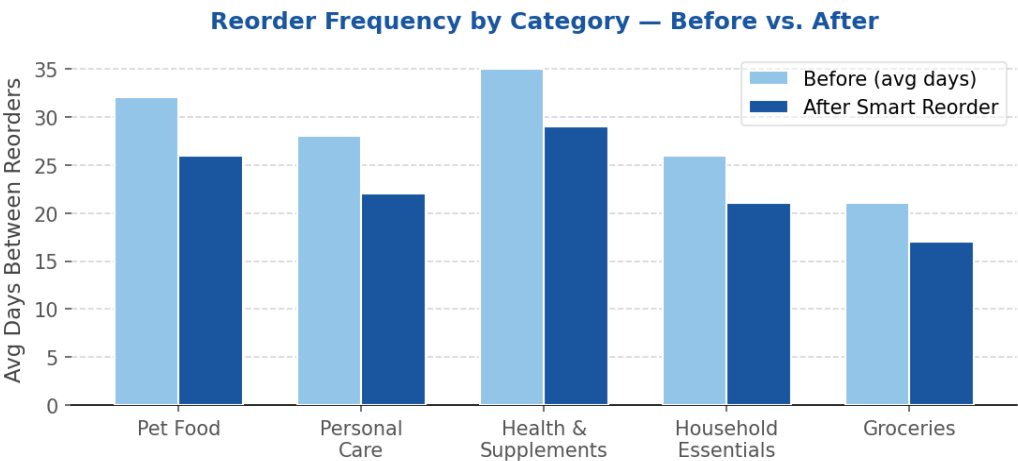


Fig. 3 — Projected Reduction in Days Between Reorders by Product Category

8. Technical Considerations

The prediction engine will leverage purchase interval modelling using a Poisson process for each user-product pair. A minimum data threshold of 3 purchases of the same SKU is required to activate predictions. The model will be retrained weekly on order history deltas and nudges will be delivered via the existing push notification infrastructure (FCM/APNS).

Dependency	Team Owner	Risk Level
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ML prediction pipeline	Data Science	Medium — new model build
Push notification service	Mobile Platform	Low — existing infrastructure
Order history API	Backend	Low — data available
My Regulars UI tab	Frontend	Low — standard component pattern

9. Go-to-Market & Launch Plan

Phase	Timeline	Scope	Success Criteria
Alpha	Month 1–2	Internal + 500 power buyers	Validate prediction accuracy ≥80%
Beta	Month 3	5% of active buyer base (A/B)	Nudge CTR >22%; reorder lift visible
GA Launch	Month 4	All repeat buyers (3+ purchases)	Full rollout + monitoring dashboard
Iteration	Month 5–6	Feature refinement from data	Optimise CTR and opt-in rate

10. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Prediction accuracy below threshold	Medium	High	80%+ accuracy bar for GA; delay rollout if not met
Notification fatigue / opt-outs	Medium	Medium	Cap 1 nudge/item/cycle; prominent snooze option
Low model coverage for new users	High	Low	Show manual reorder list until 3-purchase threshold is hit
Data privacy concerns	Low	High	On-platform only; explicit opt-in with transparent copy

11. Open Questions

- Should prediction be opt-in or opt-out by default? (Recommend opt-in for trust; revisit post-Beta data)
- How do we handle products with high price volatility at reorder time?
- What is the minimum nudge-to-purchase window that avoids last-minute UX pressure?



References: User interview synthesis (March 2026) · Competitor analysis: Amazon Subscribe & Save, Swiggy Instamart · Internal: repeat buyer cohort analysis FY2025